

Statement Purpose:

- Introduction to Wireshark and simple packet capturing.

Activity outcomes:

- After performing this lab, students will be able to understand basics of Wireshark Packet sniffer and will see “protocols in action”

1) Stage J (Journey)

One’s understanding of network protocols can often be greatly deepened by “seeing protocols in action” and by “playing around with protocols” – observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences. This can be done in simulated scenarios or in a “real” network environment such as the Internet. In the Wireshark labs you’ll be doing in this course, you’ll be running various network applications in different scenarios using your own computer (or you can borrow a friends; let me know if you don’t have access to a computer where you can install/run Wireshark). You’ll observe the network protocols in your computer “in action,” interacting and exchanging messages with protocol entities executing elsewhere in the Internet. Thus, you and your computer will be an integral part of these “live” labs. You’ll observe, and you’ll learn, by doing

2) Stage a₁ (apply)

Getting Wireshark

In order to run Wireshark, you will need to have access to a computer that supports both Wireshark and the *libpcap* or *WinPCap* packet capture library. The *libpcap* software will be installed for you, if it is not installed within your operating system, when you install Wireshark. See <http://www.wireshark.org/download.html> for a list of supported operating systems and download sites

Download and install the Wireshark software:

- Go to <http://www.wireshark.org/download.html> and download and install the Wireshark binary for your computer.

The Wireshark FAQ has a number of helpful hints and interesting tidbits of information, particularly if you have trouble installing or running Wireshark.

Running Wireshark

When you run the Wireshark program, you’ll get a startup screen, as shown below:

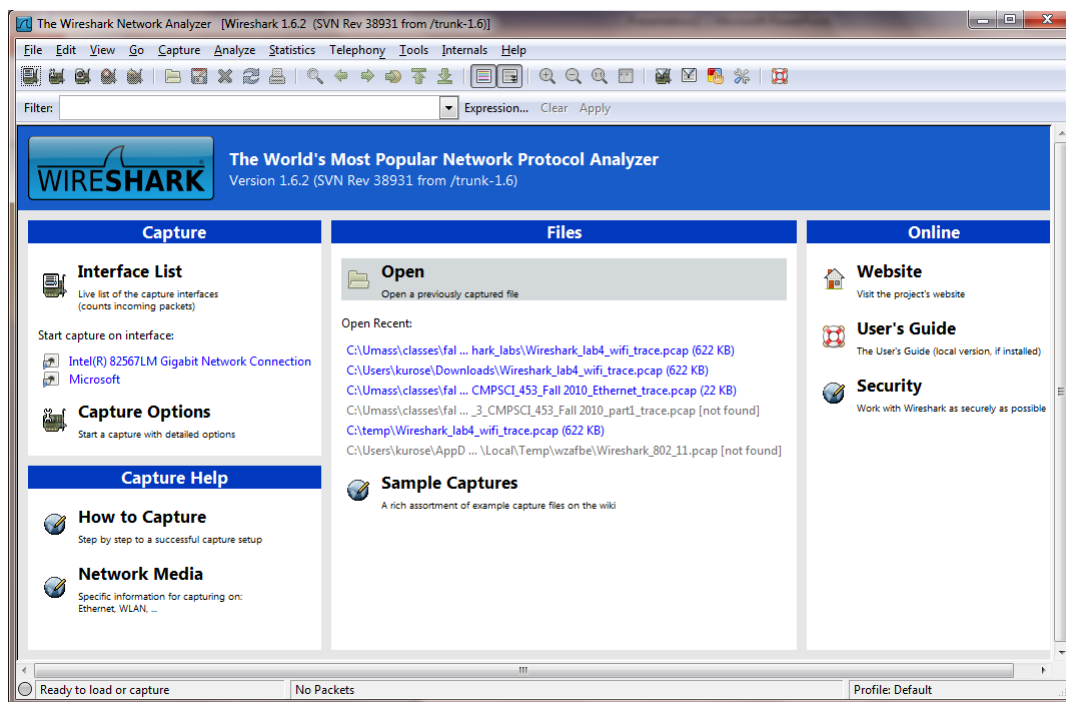


Figure 2: Initial Wireshark Screen

Take a look at the upper left hand side of the screen – you’ll see an “Interface list”. This is the list of network interfaces on your computer. Once you choose an interface, Wireshark will capture all packets on that interface. In the example above, there is an Ethernet interface (Gigabit network Connection) and a wireless interface (“Microsoft”).

If you click on one of these interfaces to start packet capture (i.e., for Wireshark to begin capturing all packets being sent to/from that interface), a screen like the one below will be displayed, showing information about the packets being captured. Once you start packet capture, you can stop it by using the Capture pull down menu and selecting Stop.

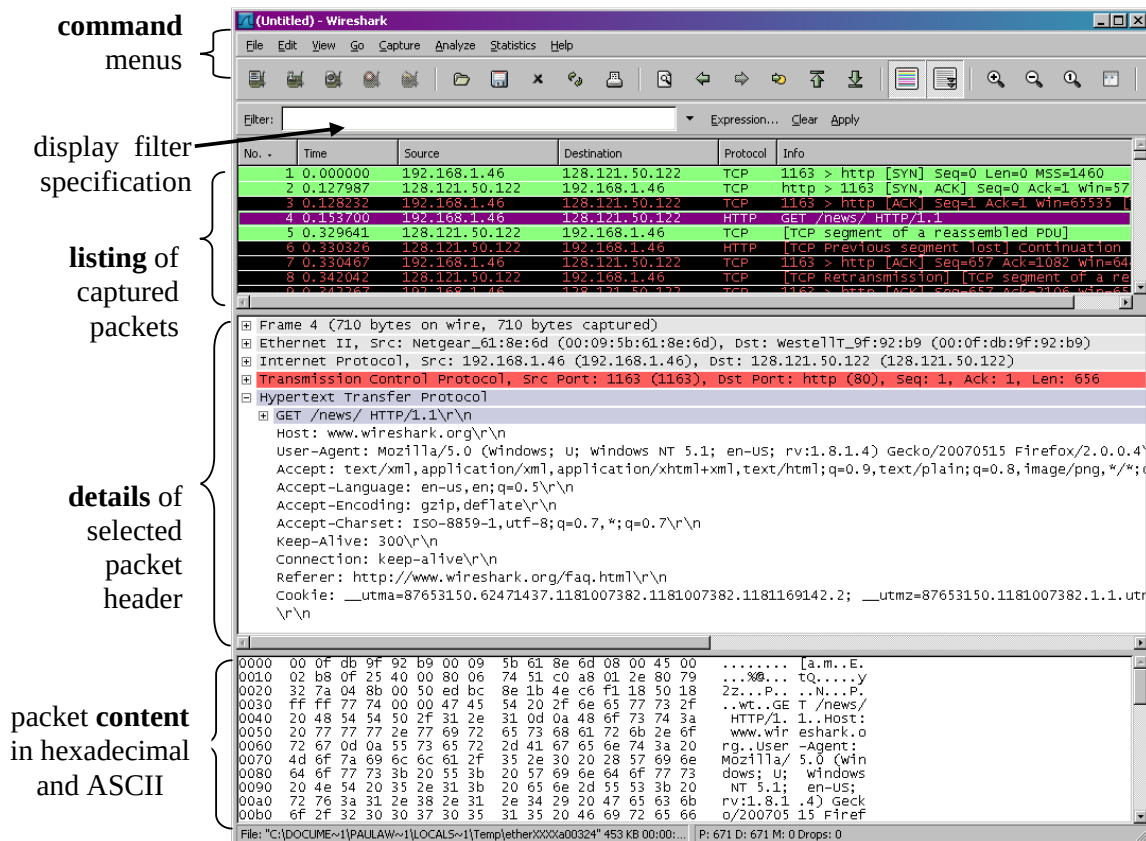


Figure 3: Wireshark Graphical User Interface, during packet capture and analysis

The Wireshark interface has five major components:

- The **command menus** are standard pulldown menus located at the top of the window. Of interest to us now are the File and Capture menus. The File menu allows you to save captured packet data or open a file containing previously captured packet data, and exit the Wireshark application. The Capture menu allows you to begin packet capture.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number (assigned by Wireshark; this is *not* a packet number contained in any protocol's header), the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet. The packet listing can be sorted according to any of these categories by clicking on a column name. The protocol type field lists the highest-level protocol that sent or received this packet, i.e., the protocol that is the source or ultimate sink for this packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. (To select a packet in the packet-listing window, place the cursor over the packet's one-line summary in the packet-listing window and click with the left mouse button.). These details include information about the Ethernet frame (assuming the packet was sent/received over an Ethernet interface) and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can

similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.

- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

Towards the top of the Wireshark graphical user interface, is the **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window (and hence the packet-header and packet-contents windows). In the example below, we'll use the packet-display filter field to have Wireshark hide (not display) packets except those that correspond to HTTP messages.

Activity 1:

The best way to learn about any new piece of software is to try it out! We'll assume that your computer is connected to the Internet via a wired Ethernet interface. Indeed, I recommend that you do this first lab on a computer that has a wired Ethernet connection, rather than just a wireless connection. Do the following

1. Start up your favorite web browser, which will display your selected homepage.
2. Start up the Wireshark software. You will initially see a window similar to that shown in Figure 2. Wireshark has not yet begun capturing packets.
3. To begin packet capture, select the Capture pull down menu and select *Interfaces*. This will cause the "Wireshark: Capture Interfaces" window to be displayed, as shown in Figure 4.

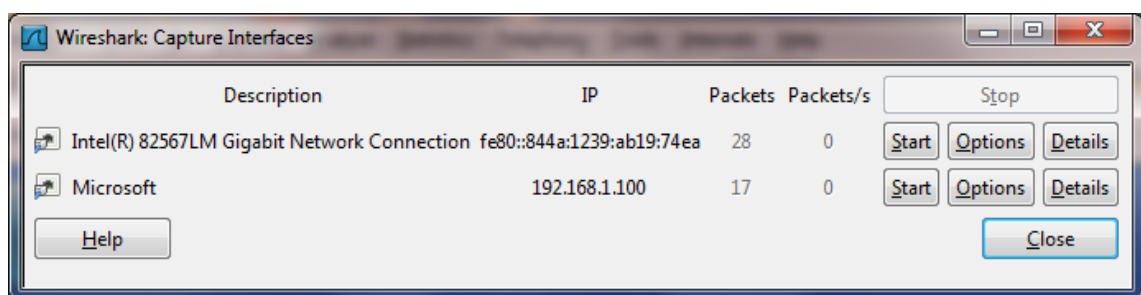


Figure 4: Wireshark Capture Interface Window

4. You'll see a list of the interfaces on your computer as well as a count of the packets that have been observed on that interface so far. Click on *Start* for the interface on which you want to begin packet capture (in the case, the Gigabit network Connection). Packet capture will now begin - Wireshark is now capturing all packets being sent/received from/by your computer!
5. Once you begin packet capture, a window similar to that shown in Figure 3 will appear. This window shows the packets being captured. By selecting *Capture* pulldown menu and selecting *Stop*, you can stop packet capture. But don't stop packet capture yet. Let's capture some interesting packets first. To do so, we'll need

to generate some network traffic. Let's do so using a web browser, which will use the HTTP protocol that we will study in detail in class to download content from a website.

6. While Wireshark is running, enter the URL: <http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html> and have that page displayed in your browser. In order to display this page, your browser will contact the HTTP server at `gaia.cs.umass.edu` and exchange HTTP messages with the server in order to download this page, as discussed in section 2.2 of the text. The Ethernet frames containing these HTTP messages (as well as all other frames passing through your Ethernet adapter) will be captured by Wireshark.
7. After your browser has displayed the `INTRO-wireshark-file1.html` page (it is a simple one line of congratulations), stop Wireshark packet capture by selecting stop in the Wireshark capture window. The main Wireshark window should now look similar to Figure 3. You now have live packet data that contains all protocol messages exchanged between your computer and other network entities! The HTTP message exchanges with the `gaia.cs.umass.edu` web server should appear somewhere in the listing of packets captured. But there will be many other types of packets displayed as well (see, e.g., the many different protocol types shown in the *Protocol* column in Figure 3). Even though the only action you took was to download a web page, there were evidently many other protocols running on your computer that are unseen by the user. We'll learn much more about these protocols as we progress through the text! For now, you should just be aware that there is often much more going on than "meet's the eye"!
8. Type in "http" (without the quotes, and in lower case – all protocol names are in lower case in Wireshark) into the display filter specification window at the top of the main Wireshark window. Then select *Apply* (to the right of where you entered "http"). This will cause only HTTP message to be displayed in the packet-listing window.
9. Find the HTTP GET message that was sent from your computer to the `gaia.cs.umass.edu` HTTP server. (Look for an HTTP GET message in the "listing of captured packets" portion of the Wireshark window (see Figure 3) that shows "GET" followed by the `gaia.cs.umass.edu` URL that you entered. When you select the HTTP GET message, the Ethernet frame, IP datagram, TCP segment, and HTTP message header information will be displayed in the packet-header window¹. By clicking on '+' and '-' right-pointing and down-pointing arrowheads to the left side of the packet details window, *minimize* the amount of Frame, Ethernet, Internet Protocol, and Transmission Control Protocol information displayed. *Maximize* the amount information displayed about the HTTP protocol. Your Wireshark display should now look roughly (Note, in particular, the minimized amount of protocol information for

¹ Recall that the HTTP GET message that is sent to the `gaia.cs.umass.edu` web server is contained within a TCP segment, which is contained (encapsulated) in an IP datagram, which is encapsulated in an Ethernet frame. If this process of encapsulation isn't quite clear yet, review section 1.5 in the text

all protocols except HTTP, and the maximized amount of protocol information for HTTP in the packet-header window).

The goal of this first lab was primarily to introduce you to Wireshark. The following questions will demonstrate that you've been able to get Wireshark up and running, and have explored some of its capabilities. Answer the following questions, based on your Wireshark experimentation:

1. List 3 different protocols that appear in the protocol column in the unfiltered packet-listing window in step 7 above.
2. How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received? (By default, the value of the Time column in the packet-listing window is the amount of time, in seconds, since Wireshark tracing began. To display the Time field in time-of-day format, select the Wireshark *View* pull down menu, then select *Time Display Format*, then select *Time-of-day*.)
3. What is the Internet address of the gaia.cs.umass.edu (also known as www-net.cs.umass.edu)? What is the Internet address of your computer?
4. Print the two HTTP messages (GET and OK) referred to in question 2 above. To do so, select *Print* from the Wireshark *File* command menu, and select the "*Selected Packet Only*" and "*Print as displayed*" radial buttons, and then click OK.

Solution:

Ans1: The following protocols appeared in the protocol column in the unfiltered packet listing window after downloading a webpage: TCP, UDP, HTTP, DNS.

Ans2: If we look at the frame section of the GET request we see that the time the packet arrived is 11:43:13.422848000

Frame 109(492 bytes on wire, 492 bytes captured)
Arrival Time: Sep 17, 2004 11:43:13.42284800
Time delta from previous packet: 6.826032000 seconds
Time since reference or first frame: 9.263432000 seconds
Frame Number: 109
Packet Length: 492 bytes
Capture Length: 492 bytes

The same section for the HTTP OK shows an arrival time of 11:43:13.43960400

Frame 110(444 bytes on wire, 444 bytes captured)
Arrival Time: sep 17, 2004 11:43:13.439604000
Time delta from previous packet: 0.016756000 seconds
Time since reference or first frame: 9.280188000 seconds
Frame Number: 110
Packet Length: 444 bytes
Capture Length: 444 bytes

The difference of these 2 times gives $.43960400 - .426032000 = 0.013572$ seconds

Ans3: If we look at the IP section of the GET request, the source and destination

Source: (128.238.244.28 (128.238.244.28)

Destination: 128.119.245.12 (128.119.245.12)

The source is the local machine's address and the destination is the web server's public My (local machine's) address = 128.238.244.28

IP address 128.119.245.12 = www-net.cs.umass.edu.

Ans4:

HTTP GET:

```
Frame 4 (862 bytes on wire, 862 bytes captured)
Ethernet II, Src: Netgear_61:8e:6d (00:09:5b:61:8e:6d), Dst: WestellT_9f:92:b9
(00:0f:db:9f:92:b9)
Internet Protocol, Src: 192.168.1.46 (192.168.1.46), Dst: 128.119.245.12
(128.119.245.12)
Transmission Control Protocol, Src Port: 1474 (1474), Dst Port: http (80), Seq: 1,
Ack: 1, Len: 808
Hypertext Transfer Protocol
GET /wireshark-labs/INTRO-wireshark-file1.html HTTP/1.1\r\n
Host: gaia.cs.umass.edu\r\n
User-Agent: Mozilla/5.0 (Windows; U; Windows NT 5.1; en-US; rv:1.8.1.4)
Gecko/20070515 Firefox/2.0.0.4\r\n
Accept:
text/xml,application/xml,application/xhtml+xml,text/html;q=0.9,text/plain;q=0.8,im
age/png,*/*;q=0.5\r\n
Accept-Language: en-us,en;q=0.5\r\n
Accept-Encoding: gzip,deflate\r\n
Accept-Charset: ISO-8859-1,utf-8;q=0.7,*,q=0.7\r\n
Keep-Alive: 300\r\n
Connection: keep-alive\r\n
```

HTTP OK:

```
Frame 6 (439 bytes on wire, 439 bytes captured)
Ethernet II, Src: WestellT_9f:92:b9 (00:0f:db:9f:92:b9), Dst: Netgear_61:8e:6d
(00:09:5b:61:8e:6d)
Internet Protocol, Src: 128.119.245.12 (128.119.245.12), Dst: 192.168.1.46
(192.168.1.46)
Transmission Control Protocol, Src Port: http (80), Dst Port: 1474 (1474), Seq: 1,
Ack: 809, Len: 385
Hypertext Transfer Protocol
HTTP/1.1 200 OK\r\n
Date: Thu, 07 Jun 2007 18:09:01 GMT\r\n
Server: Apache/2.0.52 (CentOS)\r\n
Last-Modified: Thu, 07 Jun 2007 18:08:01 GMT\r\n
ETag: "d6c69-50-cb94a240"\r\n
Accept-Ranges: bytes\r\n
Content-Length: 80
Keep-Alive: timeout=10, max=100\r\n
Connection: Keep-Alive\r\n
Content-Type: text/html; charset=ISO-8859-1\r\n
```

1) Stage **v** (verify)

Home Activities:

Activity 1:

Start a new capture, and do some arbitrary web browsing, such as visiting some Wikipedia articles, reading some news, etc. Note down all the protocols used at each layer.

2) Stage **a**₂ (assess)

Assignment#1

Note down all the protocols used at each layer.